

CORE102 Lab 3 Report

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This lab focused on capturing and analysing network traffic using tcpdump and Wireshark. I created a working directory and generated controlled ICMP, DNS, HTTP and SSH traffic.

For the HTTP exercise I started a Python web server on 127.0.0.1:8080, captured the traffic on the loopback interface, and requested the page with curl. The capture clearly showed the TCP three-way handshake (frames 1–3), the HTTP GET request (frame 4) and the 200 OK response containing the authorised test page (frame 8). The connection closed cleanly with FIN/ACK packets.

A capture filter is applied while packets are being recorded and decides what is saved to the PCAP file.

A display filter is applied afterwards in Wireshark and only controls what is shown on screen.

Because only one machine was available, a live SSH capture to a second VM was not possible. In a normal SSH session the TCP handshake is visible, but all data after that is encrypted, so passwords and commands cannot be read.

The traffic timeline confirmed that the HTTP exchange completed normally in about 0.025 seconds.

Pairing of request and response was possible using the TCP stream index and sequence numbers.

Reflection

Capture filters limit what is saved; display filters only hide packets from view. Request-response pairs are linked by stream index and sequence numbers. Capturing unrelated traffic risks exposing private information.